

● MEDIUM

● EXPRESSION OF IDEAS

Expression of Ideas

01

10 questions · ~15 min

Q1

EXPRESSION OF IDEAS

In 1873, Spanish scientist Santiago Ramón y Cajal observed that brain fibers have distinct boundaries with clear end points, a finding that went against earlier assumptions about the brain. _____ blank scientists had assumed that the brain was a continuous web of fused fibers, not a vast network of distinct, individual cells.

Which choice completes the text with the most logical transition?

- A. However,
- B. Previously,
- C. As a result,
- D. Likewise,

While researching a topic, a student has taken the following notes:

- Minnesota defines a lake as an inland body of water of at least 10 acres.
- Wisconsin's definition of a lake doesn't take size into account.
- By its own definition, Wisconsin has over 15,000 lakes, many smaller than 10 acres.
- By Minnesota's definition, Wisconsin has only about 6,000 lakes.

The student wants to contrast Minnesota's definition of a lake with Wisconsin's. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Wisconsin, which doesn't take size into account in defining a lake, claims that it has over 15,000 lakes.
- B. Because its definition of a lake is different from Minnesota's, it is unclear how many lakes Wisconsin really has.
- C. According to Minnesota's definition of a lake—an inland body of water of at least 10 acres—Wisconsin has about 6,000 lakes.
- D. Minnesota's definition of a lake—an inland body of water of at least 10 acres—is more restrictive than Wisconsin's, which doesn't take size into account.

In her 2012 analysis of tree rings from Japan’s Yaku Island, cosmic ray physicist Fusa Miyake noted an anomalous carbon-14 spike dating to 774–775 CE, indicating that a massive burst of radiation reached Earth during that time. _____blank this unprecedented radiocarbon surge was dubbed a “Miyake event” in honor of its discoverer.

Which choice completes the text with the most logical transition?

- A. Fittingly,
- B. Similarly,
- C. However,
- D. In other words,

While researching a topic, a student has taken the following notes:

- Started in 1925, the Scripps National Spelling Bee is a US-based spelling competition.
- The words used in the competition have diverse linguistic origins.
- In 2008, Sameer Mishra won by correctly spelling the word “guerdon.”
- “Guerdon” derives from the Anglo-French word “guerdun.”
- In 2009, Kavya Shivashankar won by correctly spelling the word “Laodicean.”
- “Laodicean” derives from the ancient Greek word “Laodíkeia.”

The student wants to emphasize a difference in the origins of the two words. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. “Guerdon,” the final word of the 2008 Scripps National Spelling Bee, is of Anglo-French origin, while the following year’s final word, “Laodicean,” derives from ancient Greek.
- B. In 2008, Sameer Mishra won the Scripps National Spelling Bee by correctly spelling the word “guerdon”; however, the following year, Kavya Shivashankar won based on spelling the word “Laodicean.”
- C. Kavya Shivashankar won the 2009 Scripps National Spelling Bee by correctly spelling “Laodicean,” which derives from the ancient Greek word “Laodíkeia.”
- D. The Scripps National Spelling Bee uses words from diverse linguistic origins, such as “guerdon” and “Laodicean.”

While studying jorō spiders, a large species originally from East Asia, University of Georgia researchers wondered if the spiders' rapid spread throughout the southeastern US was a result of aggressive behavior. _____blank they discovered that jorō spiders are gentle giants who react to even minor disturbances by "freezing" in place for an hour or more.

Which choice completes the text with the most logical transition?

- A. Therefore,
- B. Instead,
- C. For example,
- D. In other words,

While researching a topic, a student has taken the following notes:

- In 1965, Yale University historians claimed that a world map called the Vinland Map was drawn in the fifteenth century.
- Since that time, the map's age has been the subject of debate.
- In 2021, researchers conducted a study to analyze the elemental composition of the map's ink.
- Their analysis revealed that the ink contains a titanium compound not used in inks until the 1920s.
- The researchers concluded that the map was drawn in the twentieth century.

The student wants to present the study and its findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Given the debate about the Vinland Map's age, researchers in 2021 conducted a study to analyze the elemental composition of the map's ink.
- B. A 2021 study of the Vinland Map's ink revealed that it contains a titanium compound not used in inks until the 1920s, indicating that the map was drawn in the twentieth century.
- C. The Vinland Map, believed by some to have been drawn in the fifteenth century, was the focus of a 2021 study.
- D. Aware that a certain titanium compound was not used in inks until the 1920s, researchers in 2021 studied the elemental composition of the Vinland Map's ink.

The Coastal Futures Conservatory in Virginia is known for creating aural representations of ecological data. One such effort combines underwater audio recorded in seagrass beds with data that track rising carbon levels in the seagrass. As carbon levels increase, the audio is correspondingly distorted; _____ blank listeners can “hear” the changes in the carbon levels.

Which choice completes the text with the most logical transition?

- A. furthermore,
- B. by comparison,
- C. for instance,
- D. thus,

While researching a topic, a student has taken the following notes:

- Wool is a natural—and economically important—fiber that is obtained from animals like sheep.
- Australia is a leading producer of wool.
- The thickness of wool fibers varies across sheep breeds.
- Merino sheep produce fine wool that is used for apparel.
- Rambouillet sheep produce fine wool that is used for apparel.
- Romney sheep produce thick wool that is used for rugs and blankets.

The student wants to emphasize how Romney wool differs from Merino and Rambouillet wool. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Romney wool is just one of the many kinds of wools, each originating from a different breed of sheep.
- B. Sheep wool varies from breed to breed, so Romney wool will be different than other kinds of wool.
- C. The fine wool produced by Merino and Rambouillet sheep is used for apparel, whereas the thicker wool of Romney sheep is used in rugs and blankets.
- D. Wool is an economically important fiber—especially in Australia—that can be used to make apparel or even rugs and blankets.

Voting members of the 2002 Latin Grammys were impressed by Banda Cuisillos's album *Puras Rancheras Con Cuisillos* and its contribution to the banda genre, a form of regional Mexican music featuring large ensembles of wind instruments and drums that first developed in southern and central Mexico in the mid-nineteenth century. _____ blank they awarded the group the Best Banda Album award.

Which choice completes the text with the most logical transition?

- A. In contrast,
- B. Meanwhile,
- C. Nevertheless,
- D. Accordingly,

While researching a topic, a student has taken the following notes:

- Gravitational waves are powerful ripples that originate in deep space and eventually pass through Earth.
- The Laser Interferometer Gravitational Wave Observatory (LIGO) is a physics study that began in 2002.
- LIGO's goal is to detect and analyze gravitational waves.
- LIGO uses a pair of massive gravitational wave detectors called interferometers that are thousands of miles apart.
- In 2015, for the first time in history, LIGO researchers detected a gravitational wave passing through Earth.

The student wants to present LIGO's aim and methodology. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. In 2015, LIGO's massive interferometers detected a powerful ripple that originated in deep space and eventually passed through Earth.
- B. Though the physics study LIGO began in 2002, its massive interferometers didn't detect a gravitational wave until 2015.
- C. To achieve its aims, LIGO uses a pair of massive interferometers that are thousands of miles apart.
- D. A physics study designed to detect and analyze gravitational waves, LIGO uses a pair of massive interferometers that are thousands of miles apart.